Credit Scoring Fairness Audit

A Spark and Fairlearn case study with two million synthetic applications

Kundhan
SRN: PES1UG24AM327

What the audit found

The validation-selected policy achieved 78.02% test accuracy with an equal opportunity gap of 0.73 pp. The baseline gap was 18.29 pp at 77.49% accuracy. I used equal opportunity as the main fairness check because it measures whether applicants who would repay have similar chances of approval across groups.

Test comparison	Accuracy	TPR gap	EOdds gap	Default among approved
Baseline	77.49%	18.29 pp	30.22 pp	20.61%
Reweighing	77.49%	17.93 pp	29.79 pp	20.68%
Fairlearn thresholds	78.17%	4.94 pp	7.64 pp	19.52%
Selected policy	78.02%	0.73 pp	1.14 pp	19.89%

Each family representative maximizes validation accuracy within its family. The selected policy also meets the stated validation fairness/approval criteria. A repeated row means the same policy won both comparisons. pp = percentage points.

The selected policy is **to_tol0.010**. Its test accuracy changes by +0.53 percentage points relative to the baseline. Its overall approval rate is 66.44%, and 19.89% of approved applicants would default in this synthetic experiment.

The main lesson is that a smaller repayment-opportunity gap does not settle every lending concern. The false positive rate and the default rate among approved applicants can move differently, so I report them alongside the fairness target.

Scope: the assignment permits a large synthetic equivalent. No Home Credit records or real customer data were used. The experiment demonstrates the pipeline and its trade-offs; it does not validate a bank model.

1. Data and experimental setup

I generated 2,000,000 independent application IDs directly in Spark. Group 1 accounts for approximately 45% of applications. The two groups are fictional audit groups; they do not represent a particular gender, caste or ethnicity. Feature shifts and a biased credit-history measurement make the fairness question visible without needing a Kaggle account.

Input	Construction / role
income_z	Standard normal minus 0.35 for group 1
debt_z	Standard normal plus 0.25 for group 1
history_measured	True history minus 0.90 for group 1
employment_z	Standard normal minus 0.15 for group 1
group	Binary audit attribute; excluded from logistic model features
label	1 = would repay; 0 = would default. Generated for every applicant.

The repayment probability is sigmoid(1.1 + income_z - 1.2 debt_z + 0.9 history_true + 0.45 employment_z). The outcome is a Bernoulli draw from that probability. The model sees measured history, while the generator uses true history. This deliberately creates measurement bias. The groups also differ in their feature and repayment distributions.

Split	Applications	Purpose
train	1,200,876	Fit models and reweighing weights
validation	399,890	Fit thresholds and choose the policy
test	399,234	Final performance and fairness audit

All splits use an ID hash with seed 327, approximately 60/20/20. Separate hash salts control outcomes, calibration sampling and randomized decisions. The generator is invariant to partition count. No applicant ID or group variable enters the base model.

I fit three Spark logistic models with the same four features, L2 regularization 0.01 and at most 40 iterations. The only training change is reweighing strength. There is no imputation or external preprocessing to fit because all synthetic inputs are complete numeric values.

This dataset is equivalent in scale and task type, not a statistical replica of Home Credit. Its synthetic distributions and known outcomes make it more convenient than a real application dataset, where outcomes for rejected applicants are usually missing.

2. Metrics and mitigation methods

A decision of 1 means approve. A true positive is an approval for someone who would repay; a false positive is an approval for someone who would default. Fixing this convention matters: using default as the positive label would reverse the lending interpretation.

Metric	Definition	Why report it
Accuracy	(TP + TN) / N	Predictive correctness; can hide group errors
Equal opportunity gap	max(TPR_g) - min(TPR_g)	Unequal approval of would-repay applicants
Equalized odds gap	max(TPR gap, FPR gap)	Checks both kinds of error disparity
Demographic parity gap	max(approval_g) - min(approval_g)	Overall allocation difference
DP ratio	min(approval_g) / max(approval_g)	Relative allocation difference
Approved-default rate	FP / (TP + FP)	Risk within the approved portfolio

Method 1 - training reweighing. I estimate w(g,y) = N_g N_y / (N N_gy) from training data only [3]. The final weight is (1 - alpha) + alpha w(g,y), with alpha equal to 0, 0.5 or 1. Spark passes it to logistic regression through weightCol. At strength 1 the weighted training labels and groups are independent; this does not guarantee equal test error rates.

Method 2 - Fairlearn threshold optimization. The already fitted baseline repayment score is passed through a small sklearn-compatible adapter. ThresholdOptimizer fits group-specific, potentially randomized rules with true_positive_rate_parity and accuracy_score [1,2]. I compare tolerances 0, 0.01, 0.025, 0.05 and 0.10 on a grid of 1,000 points. These constraints apply to calibration data, so held-out gaps need not match them exactly.

The calibration sample contains 96,021 validation applications (hard cap 100,000). Exported rules are checked against Fairlearn probabilities and evaluated in Spark. For each logistic model, common thresholds from 0.30 to 0.85 in steps of 0.05 supply additional accuracy/fairness operating points.

The selection rule maximizes full-validation accuracy among policies with a TPR gap of at most 0.02 and an approval rate between 0.2 and 0.9. The minimum-gap fallback was not needed. The test split is excluded from this choice. The 2-point target and approval bounds are transparent experiment choices, not regulatory thresholds.

3. Accuracy versus fairness

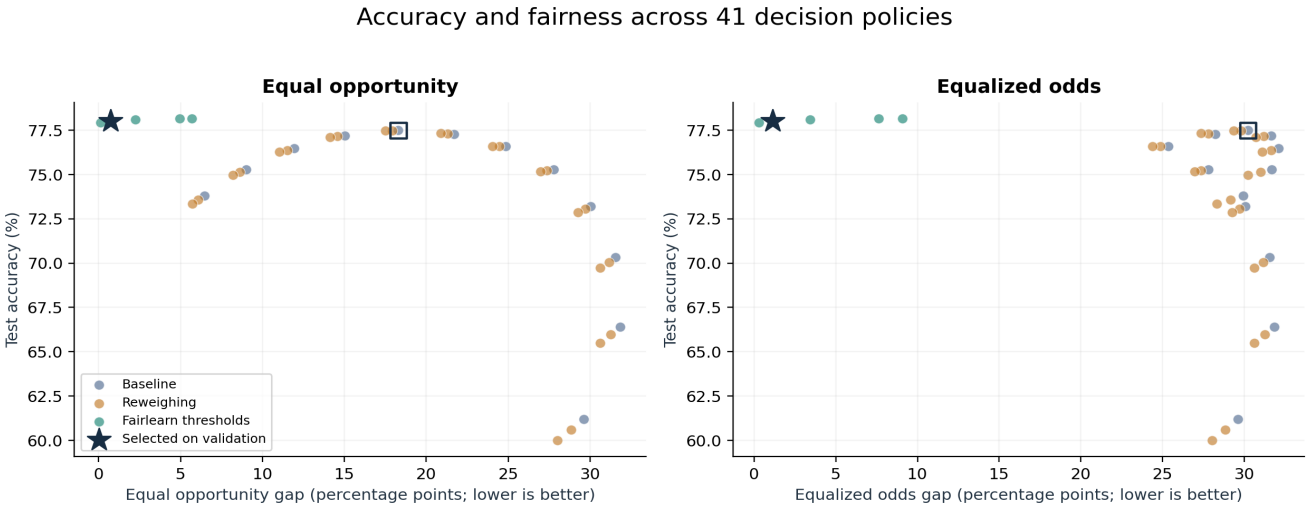


Figure 1. All 41 policies evaluated on 399,234 untouched test applications. The star is the validation-selected policy; the open square is the validation-selected baseline. These are evaluated operating points, not a fitted Pareto frontier.

The baseline TPR gap is 18.29 pp; the selected policy reduces it to 0.73 pp. The corresponding accuracy change is +0.53 percentage points. The equalized odds gap changes from 30.22 pp to 1.14 pp. The two panels matter because a policy optimized for one conditional rate can leave the other conditional rate unequal.

The best-accuracy Fairlearn family representative has 78.17% test accuracy at a 4.94 pp TPR gap. The stricter selected policy gives up 0.15 accuracy points relative to that representative. Both can outperform the baseline because the deliberately biased history measurement leaves room to improve accuracy and fairness together. Reweighing alone changes the gap from 18.29 pp to 17.93 pp; it does not directly repair that measurement bias.

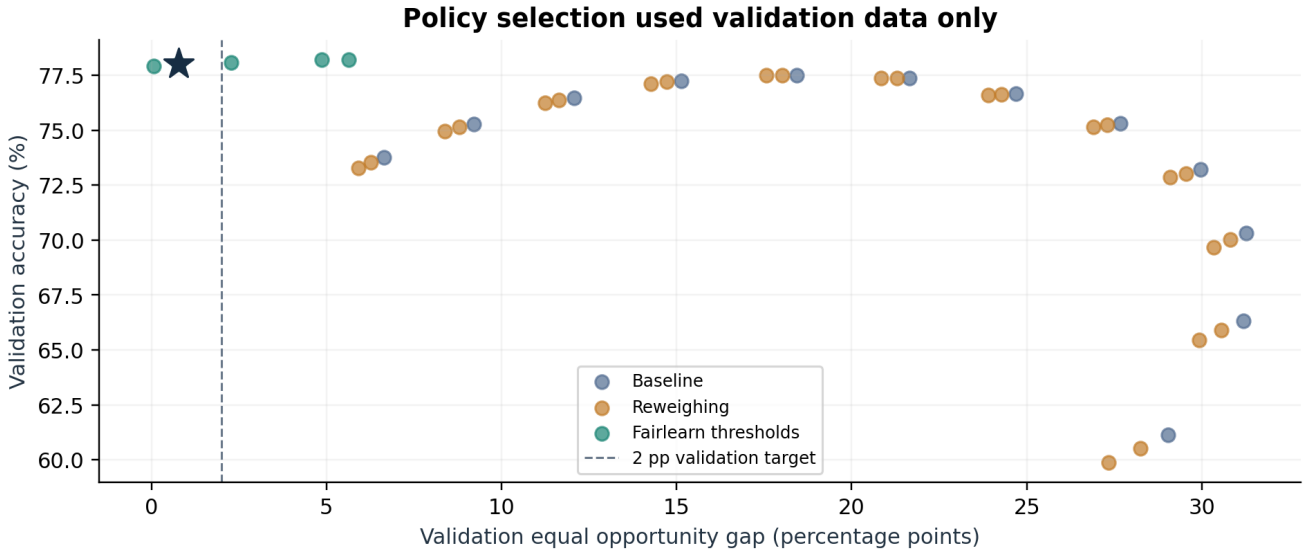


Figure 2. Validation operating points and the predeclared 2-point TPR-gap target. Only validation results determine the selected policy.

I would not choose the point furthest left without checking its approval rate and accuracy. Approving almost everyone or almost no one can make group gaps small while producing a poor lending decision rule. The approval bounds prevent those extremes from winning this experiment.

4. What changed for the two groups

Policy	Group	N	Approval	TPR	FPR	Precision
Baseline	Group 0	219,954	79.70%	93.45%	52.07%	78.30%
Baseline	Group 1	179,280	51.76%	75.16%	21.86%	81.47%
Selected	Group 0	219,954	69.28%	86.16%	35.36%	83.04%
Selected	Group 1	179,280	62.96%	85.43%	34.22%	76.14%

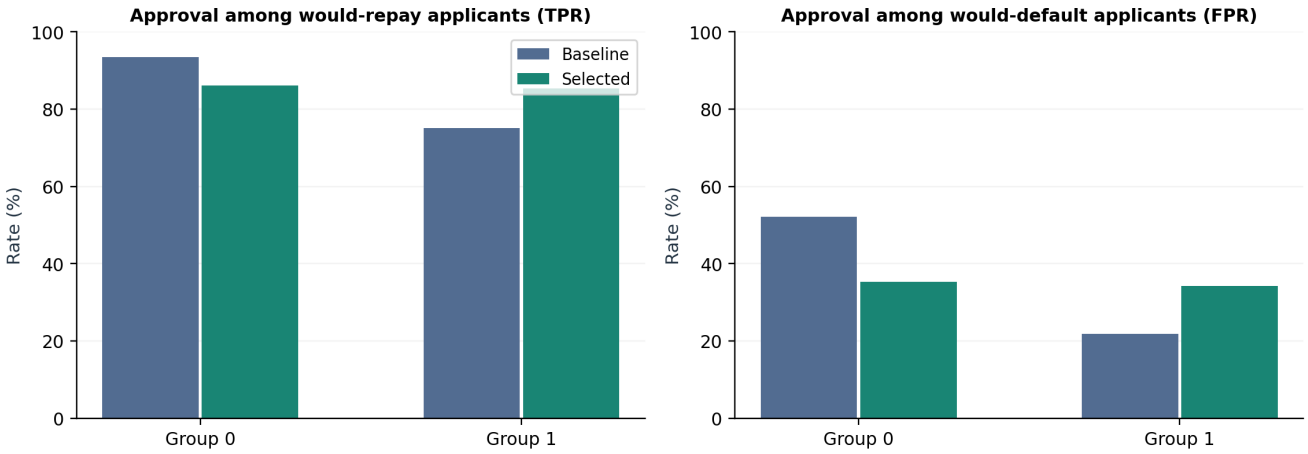


Figure 3. Approval among applicants who would repay (left) and who would default (right). A lower TPR gap is desirable, but a high FPR still matters to affordability and portfolio risk.

For the selected policy, the signed TPR difference (group 0 minus group 1) is 0.73 pp. An approximate 95% binomial interval is [0.44 pp, 1.01 pp]. This interval is conditional on the fitted policy and this synthetic population. It does not include retraining uncertainty, drift or uncertainty about the data-generating assumptions.

The selected demographic parity gap is 6.33 pp and the DP ratio is 0.909. Neither is a lending compliance pass/fail test. Different outcome base rates mean equal opportunity and equal overall approval rates need not coincide.

The threshold policy needs group membership when making a decision, while the reweighed model uses it only during training. That difference affects how each method could be governed. Even with accurate group data, a real deployment would need appropriate review of whether and how those attributes may be used.

5. Scale, checks and reproduction

The complete audit ran on 2,000,000 rows with Spark 3.5.3, local[4], and 8 partitions. The audit stage took 19.6 seconds (0.3 minutes). This is a local execution measurement, not a claim about a production cluster or linear speedup.

Stage	Where it runs / what is collected
Generate and split	Spark SQL hash expressions; all 2,000,000 rows remain distributed
Train / reweigh	Spark ML logistic regression; four group-label counts fit weights
Calibrate	At most 100,000 validation score rows collected for Fairlearn
Evaluate	Spark exports only policy-by-split-by-group confusion counts
Cross-check	Fairlearn MetricFrame uses 8 weighted cells per policy/split
Document	Small metrics CSVs produce all tables and plots

The maximum Spark/Fairlearn metric difference was 0; the maximum exported threshold-probability error was 0. Both checks assert an error below 1e-10. Weighted cells reproduce the reported confusion-based metrics exactly; this shortcut cannot substitute for raw scores when checking ranking, calibration or individual fairness.

Recorded run evidence

The following lines are taken from the actual saved run log. They are text evidence, not a screenshot of a different environment.

```
RUN rows=2,000,000 Spark=3.5.3 seed=327

TRAIN alpha=0.0 seconds=3.3

TRAIN alpha=0.5 seconds=1.3

TRAIN alpha=1.0 seconds=1.1

CALIBRATION rows=96,021; policy count=41; export max error=0.0

SELECTED to_tol0.010

VERIFIED Fairlearn max error=0.0; elapsed=19.6s
```

Reproduce or demonstrate

Open the included notebook in Colab and run the cells in order, or follow README.md for a local Java 17 / Python setup. A 20,000-row smoke run checks the full path quickly; the submitted findings come from the two-million-row run. No API key is required. The CSVs and model artifacts remain available even when a full rerun is inconvenient.

The package includes pinned dependencies, model coefficients, Spark model directories, threshold rules, group confusion counts, the calibration sample, a 1,000-row input example, run metadata and numerical checks. Input generation is reproducible, so the full synthetic table is not bundled.

6. Which metric suits lending?

For this case, I would use equal opportunity as the primary applicant-facing metric. With repayment as the positive label, it asks a practical question: among applicants who would repay, how often does each group receive approval? It directly addresses the harm of denying a viable borrower. Equalized odds is a useful stricter companion because it also checks unequal approval among applicants who would default [2,6].

I would keep false positive rate and default among approved applicants as separate risk checks. The selected policy was not optimized for loan size, interest, loss given default or affordability because the brief supplied none of these. Accuracy gives the two mistakes equal weight; a lender would need an explicit cost and affordability model before setting an operating point.

Demographic parity answers a different question: whether groups receive approvals at similar overall rates. It is useful for spotting allocation differences, but it ignores repayment outcomes. I would not use a single DP ratio, or an imported numerical cutoff, as proof that a lending system is fair.

Limits of the conclusion

The data has two fictional groups, complete labels and one controlled form of measurement bias. It does not cover intersectional groups, missing demographics, unequal observation windows, outcome censoring, temporal drift or the effect of an actual loan offer. Real lenders observe repayment mainly for approved borrowers, so naive error-rate estimates can be affected by selection bias. Simply treating rejected applicants as defaults would invalidate the audit.

I would take forward the validation-selected policy for further testing, rather than deploy it. The next checks would use representative time-based holdouts, more demographic slices, repeated training seeds, score calibration and realistic lending costs. The model card records these boundaries so that a numerical fairness improvement is not mistaken for a complete lending approval.

References

[1] Fairlearn 0.13. ThresholdOptimizer API and postprocessing guide. https://fairlearn.org/v0.13/api_reference/generated/fairlearn.postprocessing.ThresholdOptimizer.html

[2] Hardt, Price and Srebro (2016). Equality of Opportunity in Supervised Learning. https://arxiv.org/abs/1610.02413

[3] Kamiran and Calders (2012). Data preprocessing techniques for classification without discrimination. https://doi.org/10.1007/s10115-011-0463-8

[4] Apache Spark 3.5.3. ML classification and regression; logistic regression. https://spark.apache.org/docs/3.5.3/ml-classification-regression.html#logistic-regression

[5] Mitchell et al. (2019). Model Cards for Model Reporting. https://arxiv.org/abs/1810.03993

[6] Fairlearn. Common fairness metrics and their limitations. https://fairlearn.org/v0.13/user_guide/assessment/common_fairness_metrics.html